MILITARY SPECIFICATION

INDICATOR PLUG, FOR MISSILE PROPELLANTS

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification establishes the requirements for an indicator plug to detect by color change, leaks of certain propellants, principally mixed amine fuels and fuming nitric acid.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-00225/8	Aluminum Alloy Bar, Rod, Wire, and Special Shapes, Rolled, Drawn, or Cold Finished, 6061
UU-P-63	Paper, Blotting
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-676	Box, Set-up

Military

MIL-B-117	Bags and Sleeves, Interior Packaging
MIL-S-6855	Synthetic Rubber, Sheets, Strips, Molded or Extruded
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for

FSC 6685

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS33586	Metals, Definition of Dissimilar

(When requesting any of the applicable documents, refer to both title and number. Copies of the specifications and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 1051, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Materials - Materials used in the manufacture of the indicator plug shall conform to the applicable Government specifications or where Government specifications are not involved shall be of high quality and suitable for the intended purpose.

3.1.1 Metal - Metal component parts of the indicator plug shall be manufactured from an aluminum alloy conforming to QQ-A-00225/8. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact. Dissimilar metals are defined in MS33586.

3.1.2 Gasket material - Unless otherwise specified, rubber gasket components shall be fabricated from synthetic material conforming to MIL-S-6855, Class II, Grade 60.

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3.1.3 Spacer-separating washers - The spacer-separating washer material shall be polytetrafluoroethylene.

3.1.4 Plastic - Material used in the manufacture of the color code disk shall be a weather resistant plastic. The color shall be yellow, matching approximately Color Chip No. 23793 of FED-STD-595.

3.1.5 Transparent viewer - The viewer shall be a transparent plastic such as plexiglass UF-3, capable of absorbing or reflecting ultra-violet light.

3.1.6 Propellant leak indicator element - Material used in the manufacture of the indicating element shall be blotting paper conforming to UU-P-63, Grade B, color white, halftone, and smooth on both sides. The thickness shall be 0.0185 ±0.0035 inch. The element shall consist of two semicircles of blotting paper. One semicircle shall be treated with enough N-phenylanthranilic acid to turn bright yellow when exposed to fuming nitric acid. The other semicircle shall be treated with enough mercurous chloride to turn black when exposed to the mixed amine fuel fumes. A thin green line shall be drawn across the center of the element to denote the treated side. The indicating paper containing N-phenylanthranilic acid shall be prepared by adding several drops of a mixture of 0.4 percent or more by weight of N-phenylanthranilic and 0.4 percent of water dispersible titanium dioxide (Titanox RAF-1 or equal) in methyl alcohol. The indicating paper containing mercurous chloride shall be prepared by adding several drops of a mixture containing 15 percent or more by weight of mercurous chloride and 1 percent by weight of carboxymethylcellulose in water.

3.2 Design - The propellant leak indicator shall be so designed that the indicating element is freely exposed to a confined atmosphere such as in a drum container, missile container, or behind a bulkhead door. The design shall be such that only common hand tools are required to install and remove the plug from the container wall. Unless otherwise specified, dimensions shall be in conformance with Figures 1 and 2.

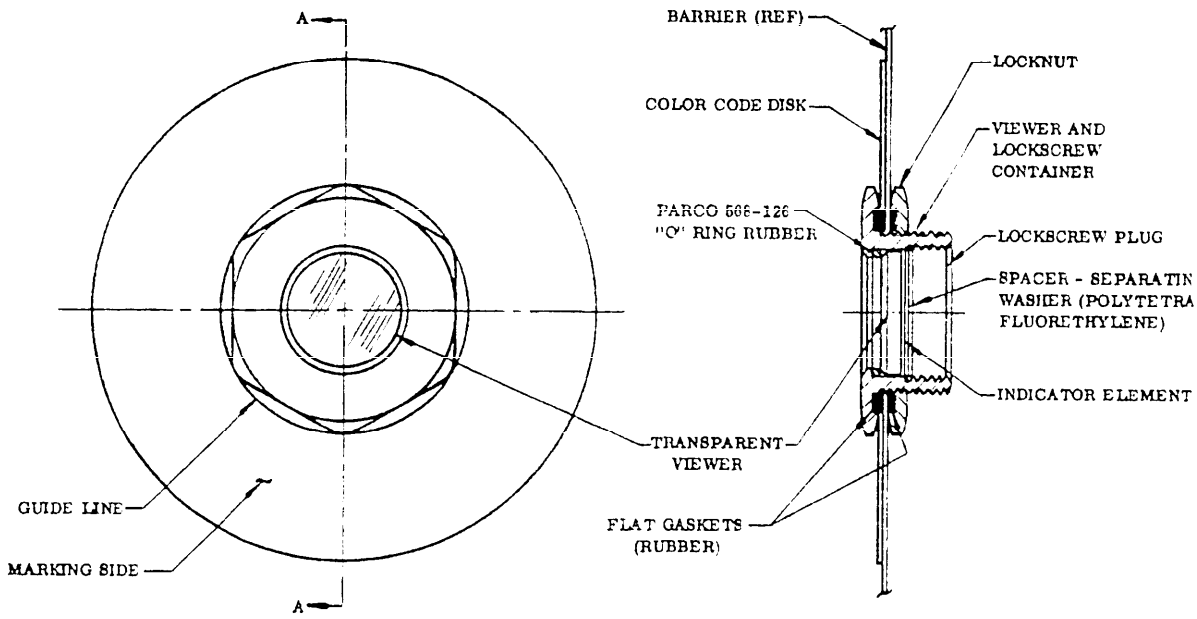
3.3 Construction and assembly - The leak indicator plug shall be so constructed that it will withstand the strains, shocks, and vibrations, due to shipping, storage, and service usage. Both the construction and assembly of the leak indicator plug shall be in accordance with Figures 1 and 2.

3.3.1 Screw threads - Unless otherwise specified, screw threads shall be 1-1/2 NPS 11-1/2 (straight pipe thread) and shall conform to Handbook H28.

3.4 Identification and markings - (See 6.2.)

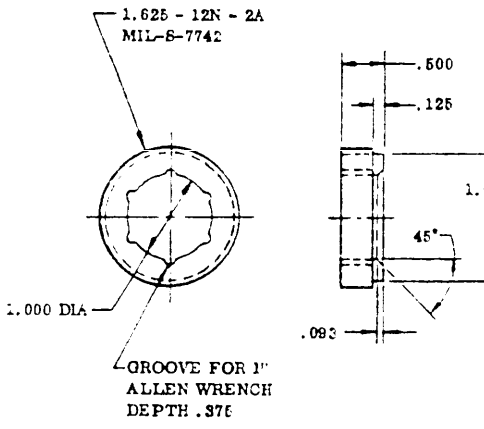
3.4.1 Color code disk - The color code disk shall be marked for identification with lettering 1/8 inch high printed directly on the disk or on

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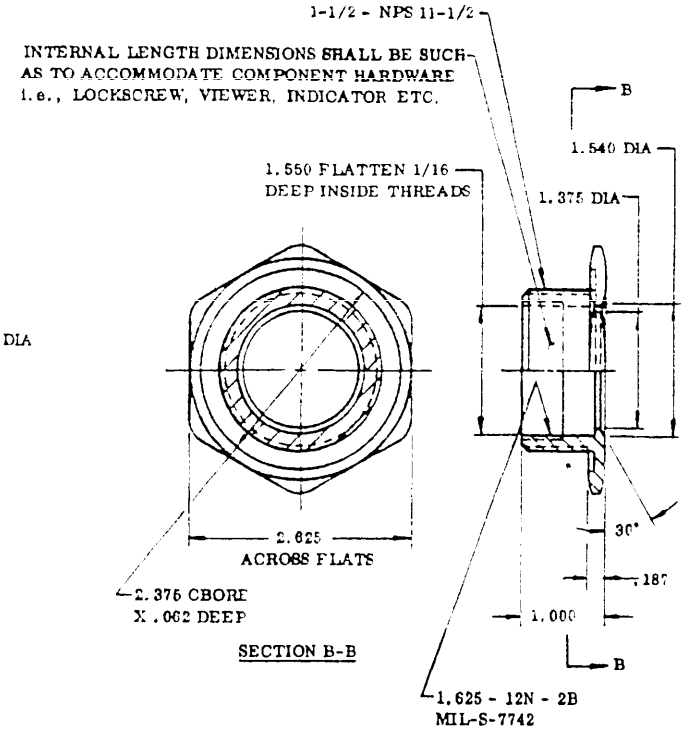


SECTION A-A

ASSEMBLY



LOCKSCREW PLUG



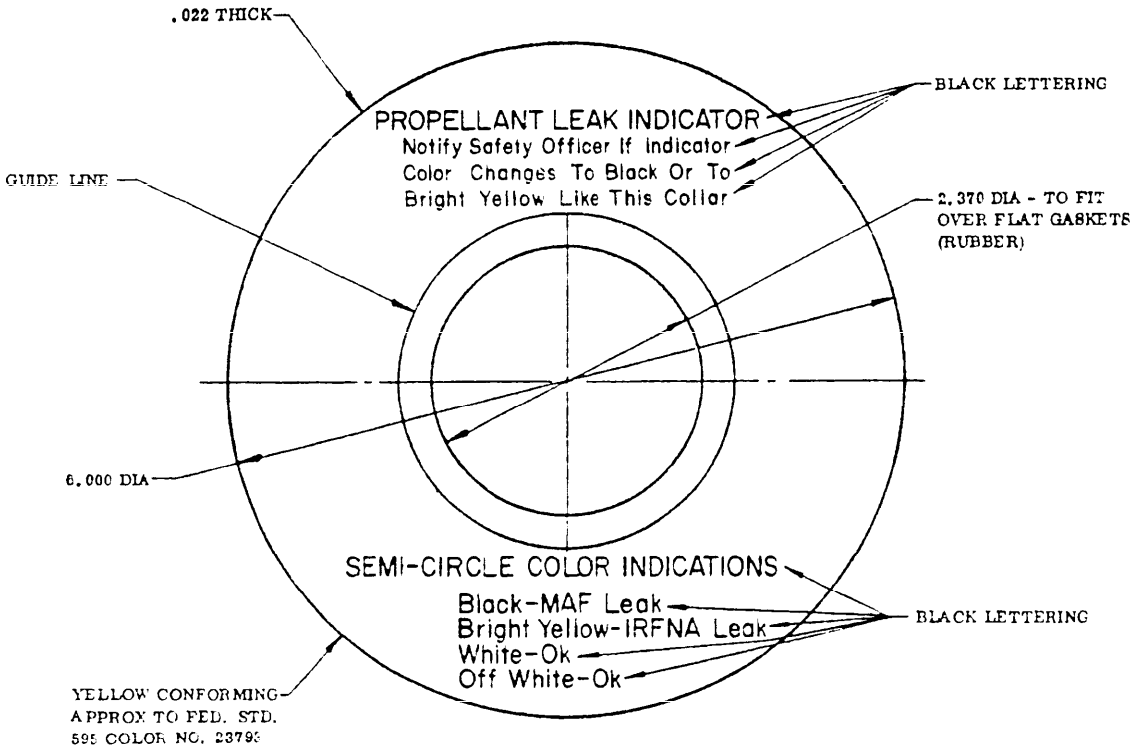
SECTION B-B

DIMENSIONS IN INCHES

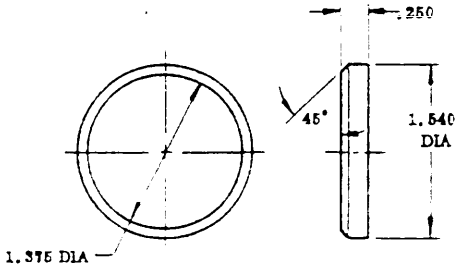
VIEWER AND LOCKSCREW CONTAINER

Figure 1

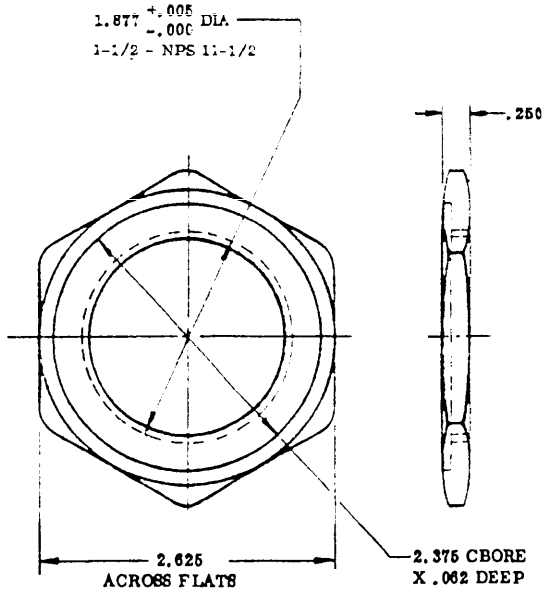
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COLOR CODE DISK



TRANSPARENT VIEWER



LOCKNUT

DIMENSIONS IN INCHES

Figure 2

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labels which shall be waterproof and weatherproof. Lettering shall not be obliterated by normal handling or immersion in water and shall not fade under exposure to direct sunlight.

3.5 Color and sensitivity of the indicating element - When the indicator plug is subjected to the test of 4.5.1, the color of one semicircle of indicating paper shall be bright yellow, and the color of the other semicircle of the indicating paper shall be black.

3.6 Simulated sunlight exposure test - When the plug is exposed to the simulated sunlight exposure of 4.5.2, there shall be no evidence of discoloration of the indicating element in the plug.

3.7 Workmanship - The complete indicator plug shall show no mold marks, cracks, splits, nor other surface or visible defects which may impair the physical properties and performance characteristics of the indicators in any way.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for inspection - Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105 except where otherwise indicated. For purposes of sampling, an inspection lot shall consist of all material made by the same process, from the same manufacturer, and to be delivered at one time.

4.2.1 Inspection of components and materials - In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.2.2 Examination of the end item - The end item shall be examined in accordance with the classification of defects, inspection levels, and acceptable quality levels (AQL'S) set forth below. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of indicators for examinations under 4.2.2.1 and 4.2.2.2, and in units of shipping containers for examination under 4.2.2.3.

4.2.2.1 Examination of the end item for defects in appearance, construction, identification and workmanship - The sample unit for this examination shall be 1 indicator plug.

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EXAMINE	DEFECTS
Material	Dissimilar metals in intimate contact
	Gasket or sealing material adjacent to indicating element
	Metals not conforming to QQ-A-00225/8
	Indicating element blotting paper material not conforming to UU-P-63, Grade B
	Rubber material not conforming to MIL-S-6855, Class II, Grade 60
	Color code disk material not conforming to 3.1.4
	Separating washer material not conforming to 3.1.3
	Viewer not transparent
Construction	Indicating element is not freely exposed to the interior surface of container
	Special tools required to assemble and disassemble
	Exterior gaskets do not seat properly in plug groove
	Blotting paper not white in color, halftone and smooth on both sides
	Indicating element not placed in plug treated side up
Identification	Illegible Incorrect Incomplete
Workmanship	Mold marks Cracks Tears Splits Surface defects Burrs

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4.2.2.2 Examination of the end item for defects of dimension - The sample unit for this examination shall be one indicator.

EXAMINE	DEFECTS
Blotting paper	Blotting paper thickness varies more than tolerance specified in 3.1.6
Color code disk	Disk thickness varies more than tolerance specified in 3.1.4
Plastic viewer	Viewing area not as specified in 3.1.5
Screw thread	Thread size not 1-1/2 NPS 11-1/2
Color code disk markings	Lettering not 1/8 inch high

4.2.2.3 Examination of preparation for delivery - Examination shall be made to determine compliance with packaging and packing and marking requirements of this specification. The sample unit shall be one shipping container fully packed and selected just prior to the closing operation.

EXAMINE	DEFECTS
Material	Packaging: Heat sealable bag does not conform to MIL-B-117, Class E
	Packing: Indicator not snugly packed in set-up box
Workmanship	Incomplete or insecure sealing of container flaps
	Quantity differs from that specified in contract or purchase order
	Packing: Insecure, loose strapping or tape banding
	Bulging or distortion of shipping container
	Broken, bruised, crushed or otherwise damaged shipping container which may result in injury to or loss of contents

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EXAMINE	DEFECTS
Workmanship (Continued)	Markings (unit package and shipping containers):
	Omitted, incomplete, incorrect, illegible
	Of improper size, location, sequence or method of application
Weight	Gross weight in excess of weight limitations of container specification

4.3 Inspection levels and acceptable quality levels for examination - The inspection levels for the purpose of determining the sample size and the acceptable quality levels (AQL'S) shall be expressed in defects per 100 units for 4.2.2.1 and 4.2.2.2, and percent defective for 4.2.2.3, as follows.

EXAMINATION PARAGRAPH	INSPECTION LEVEL	AQL
4.2.2.1	I	2.5
4.2.2.2	I	2.5
4.2.2.3	S-4	6.5

4.4 Quality conformance tests - Quality conformance tests are those tests performed on indicator plugs produced in accordance with the requirements of this specification and have been submitted for acceptance under a contract or order. (See 4.5.)

4.4.1 Samples for quality conformance test - A sample unit consisting of two indicator plugs shall be randomly selected from each inspection lot in accordance with MIL-STD-105, Inspection Level S-2 and acceptance number zero. The sample unit as referenced herein is defined as the number of indicator plugs required to perform the test.

4.5 Test methods -

4.5.1 Color and sensitivity of the indicating element -

4.5.1.1 Apparatus - The apparatus shall consist of two desiccators each containing silica gel as the desiccant. Five cubic centimeters of fuming nitric acid shall be poured in a small glass container contained in one desiccator. In the other desiccator, five cubic centimeters of mixed amine fuel shall be poured into a small glass container.

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4.5.1.2 Procedure - An indicator plug containing the indicating element shall be placed in each desiccator. The color of the indicating paper shall be checked within fifteen minutes for color change. The color of one semicircle of indicating paper shall be bright yellow. The color of the other semicircle of indicating paper shall be black.

4.5.2 Simulated sunlight exposure -

4.5.2.1 Test procedure - Place the indicator plug under a R.S. Sunlamp for 500 hours. The lamp shall be positioned 15 inches from the base of a 16-inch diameter turntable which shall revolve at 3 RPM. The temperature on the turntable shall be maintained at 120 ±2° F. A power pac shall be installed in the line to maintain a constant line voltage to the lamp. Before exposing the indicator plug under the sunlamp, the lamp shall have burned for a minimum of 100 hours. The lamp must also be in operation at least 3 hours immediately before plug exposure. After exposure, the plug shall be examined for conformance with the requirement. (See 3.6.)

4.5.3 Rejection criteria - Failure of the indicator plug to conform to the above tests shall reject the lot represented.

5. PREPARATION FOR DELIVERY

5.1 Packaging - Each indicator plug shall be packaged in a heat sealable bag conforming to MIL-B-117, Class E and snugly packed in a folding or set-up box conforming to PPP-B-566 or PPP-B-676, respectively.

5.2 Packing - Packing shall be Level A, B, or C as specified in the contract or order and in accordance with MIL-STD-794(WP). As far as practical, containers shall be of minimum tare and cube consistent with the protection required and contain identical quantities.

5.3 Marking - In addition to any special marking specified by the contract or order, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use - The indicator plug covered by this specification is intended for use in rigid wall or flexible barrier containers which contain missile sections enclosing mixed amine fuels and fuming nitric acid.

6.2 Ordering data - Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification

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(b) Quantity

(c) Special markings or identification (see 3.4)

(d) Level of packaging and level of packing required (see 5.1 and 5.2)

6.3 This specification contains the requirements of MS17868(Weps), Indicator, Propellant Leak, Plug, Color Change, for Unsymmetrical Dimethylhydrazine and Fuming Nitric Acid.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-I-81508 (AS) INDICATOR PLUG, FOR MISSILE PROPELLANTS

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED *(Printed or typed name and activity)* | DATE

DD FORM 1 NOV 64 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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